

ALEXANDER SANCHEZ CYRIL

Software Engineer — Full-Stack, Backend & AI Systems

alexander.s.cyril@gmail.com | +234 902 216 2214 | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Software engineer with 3+ years of experience shipping full-stack, backend, and AI-powered systems across independent projects and collaborative engineering teams. Author of Portfolio-Resilience, a production-grade .NET resilience library with real-world adoption, and current Open Source Engineering Fellow at ArcellAI, building provenance-aware AI systems for economic analysis. Skilled in scalable API design, authentication, distributed architectures, and agentic AI workflows across Python, TypeScript, C#, Java, and Go.

KEY ACHIEVEMENTS

- Built scalable APIs, authentication services, messaging systems, and database-driven applications in Python, TypeScript, C#, Java, and Go across Django, FastAPI, ASP.NET Core, Fastify, Gin, and Next.js
- Delivered SaaS platforms, AI-powered applications, and developer-productivity tools end to end, from architecture through deployment
- Applied distributed-systems patterns — caching, async messaging, event-driven workflows, and session management using Redis, RabbitMQ, and Kafka
- Deployed cloud-hosted applications with Docker, CI/CD pipelines, and Infrastructure as Code (Terraform) on Vercel, Render, and Neon
- Authored and published Portfolio.Resilience, a production-grade .NET 10 resilience library on NuGet with 548 automated tests and adoption in File-Ferry v1.0.0

TECHNICAL SKILLS

Category	Skills
Languages	Python · JavaScript/TypeScript · C# · Java · Go
Backend & APIs	ASP.NET Core · Spring Boot · Django · FastAPI · Express · Fastify · Gin
Frontend	React · Next.js · Tailwind CSS · HTML5/CSS3
AI & Agentic Systems	LangGraph · LangChain · LangSmith · LlamaIndex · RAG · MCP · Vector Databases · Multi-Agent Systems
Databases	PostgreSQL · MongoDB · MySQL · SQLite
Cloud & DevOps	Docker · GitHub Actions (CI/CD) · Terraform · Vercel · Render · Neon
Distributed Systems	Microservices · Event-Driven Design · REST APIs · JWT Auth · RabbitMQ · Kafka · Redis
Tools	Git · Postman · Swagger/OpenAPI · Linux

PROFESSIONAL EXPERIENCE

Open Source Engineering Fellow — ArcellAI Inc.

Sept 2026 — Present

San Francisco, CA, USA (Remote) · [arcell.ai](#)

- Contribute to VEDA (Vendor Economic Dependency Assessment), a provenance-aware AI system for vendor economic dependency and related-party analysis
- Design entity-resolution pipelines, multi-source evidence retrieval, normalization, conflict detection, and human-review workflows centered on a relationship/evidence context graph
- Contribute to federated system architecture, source connectors, and agent design, including canonical modeling, semantic layer, and ontology mapping for context-driven agents

Self-Employed Software Engineer

April 2023 – Present

End-to-end owner of architecture, backend, frontend, database design, API development, cloud deployment, and maintenance across several independent projects, summarized below.

Portfolio-Resilience — Open-Source .NET 10 Resilience Library

Author & Maintainer · NuGet: Portfolio-Resilience · GitHub: sancy1/portfolio-resilience · MIT License

- Designed, built, and shipped a production-grade .NET 10 resilience library to NuGet with zero third-party runtime dependencies
- Implemented retry with exponential backoff and jitter, circuit breaker, per-attempt timeout, a 4-strategy rate limiter, bulkhead, hedging, fallback, policy composition, structured logging, and p50/p95/p99 latency metrics with ambient correlation IDs
- Added four capabilities beyond Polly's first-class APIs: idempotency-key propagation, PCI-safe event scrubbing, total time budget, and a Payment Processing Preset (PPP) for non-idempotent writes
- Reached 548 automated tests (0 failures, 0 warnings) across Windows and Linux CI, and published 3 coordinated NuGet packages via Trusted Publishing (OIDC)
- Adopted in production by File-Ferry v1.0.0, a Windows desktop application rated 9.2/10 by users, with zero production incidents attributable to the library across 10,000+ directory walks

EL-Exousia Weather Intel — Agentic Weather Intelligence Platform

Stack: FastAPI · React · LangGraph · LangChain · LangSmith · LlamaIndex · MCP

- Built an AI-powered weather intelligence system using multi-agent orchestration and tool-calling LLM workflows in LangGraph
- Designed a retrieval-augmented generation (RAG) pipeline with vector memory for contextual and historical weather reasoning
- Integrated real-time weather APIs into the agentic pipeline to drive dynamic forecast generation
- Implemented modular, multi-step agent workflows for structured decision-making and response generation

AxioQuan — E-Learning & Marketplace Platform

Stack: Java · C# · Fastify (TypeScript) · Next.js

- Designed and built a full-stack e-learning and digital marketplace platform with course creation, quizzes, and paid content delivery
- Implemented role-based access control for students, instructors, and administrators
- Built real-time messaging between students and instructors, plus a payment flow for enrollment and content access
- Engineered the backend APIs and database schema for scalable content management and delivery

LEADERSHIP & ACADEMIC PROJECT EXPERIENCE

- Led student project teams in university software development courses, coordinating planning, task distribution, and system design decisions
- Served as technical lead in Agile-style environments, managing sprint planning and feature delivery
- Supported the team through code reviews, debugging help, and technical guidance
- Applied core engineering practices — Git workflows, testing, and system design to deliver team projects from requirements to deployment

EDUCATION

Bachelor of Science in Software Development — Brigham Young University–Idaho	2026
Associate of Applied Science in Software Development — Brigham Young University–Idaho	2026

CERTIFICATIONS

- Software Development — Brigham Young University–Idaho, 2026
- Web Development — Brigham Young University–Idaho, 2025
- Web & Computer Programming — Brigham Young University–Idaho, 2024